Part1-intro

Hello, my name is ________,

Thank you _____ for participating in our study. We aim to gather insights into testing for and measuring discriminatory behavior in the outcome of software systems. We look forward to utilizing your insights to further our research.
Now, we will review the consent form included in the interview sign-up survey.

[Revisit the consent form and open for questions]

This interview will be recorded for internal purposes only. Additionally, we will be taking notes during the interview. The consent form that was sent to you specifies that your identity will remain confidential. Other than your email address, no other personal information will be collected. Your email will be used only to schedule your interview and compensation. You can withdraw the study at any point. Are you still willing to participate in this study?

If you get disconnected from the call, please use the same link included in the email invitation, join via phone, or email if you need to reschedule.

Before we begin, do you have any questions for me?
I'll begin the interview recording now.
[Start Recording]

Background Questions

1. To begin, tell us a little about yourself.
2. What is your current role/occupation?
   a. How familiar are you with traditional software testing methodologies? Maybe you can share the kind of experience in software testing with us.

Interview Questions

Thank you for responding! Now I want to talk more about your experiences & expectations in understanding discriminatory behavior in software systems.

*********** **Part 1*****************

Section: Experiences with Discrimination in Software Outcomes

1. What is your experience in the software engineering field?
2. Have you used a Large Language Model for software engineering tasks?

3. Do you have any experience in encountering discriminatory software behavior?
4. When you recognized that potential for discriminatory behavior, what was your *expectation* of how the system should have behaved instead?

Part2-Themis

*******Part 2********

Thank you for sharing your experiences and perceptions regarding software discrimination. Next, we would like to get your perception on the utility and usability of different approaches to create unit tests for measuring and understanding discriminatory behavior in software outcomes. For this study we are considering two different test generation tools. First is Themis, an automated causal unit test generation tool. Secondly ChatGPT which has been identified as test generation tool for several domain

Now I will elaborate for you 2 separate scenarios. I want you to open our project and do the following tasks according to instructions, which you will have access to as you are working

Section 1: Themis Run

Scenario 1: This is a simple loan application system which is a Rule-based software loan.py. The behavior of this software is quite deterministic. We aim to understand the user perspective on understanding discriminatory behavior of such systems through unit test generation approaches. Similarly an ML based program which is probabilistic.

- **Program run**

A. Scenario: Given the loan approval case, participants define input attributes (e.g., gender, race, income) as categorical.

C. Through Themis generate unit tests running grid2.py for the loan program.

D. look at the results and go to the following question

E. Similarly run grid3.py and generate result for ML based program and answer the following question

- **Think-Aloud Prompt**

1. Did you face any challenges during setting up the run environment
2. What do you think these unit tests are verifying/communicating for both of the programs? Any challenges understanding them?
3. Did you face any challenge during test generation or understanding generated tests?

4. Do you think the generated unit tests help you reason about how the program shows discrimination?
    - If so, why?
    - If not, why not?

5.Based on these results, do you notice any potential next action you could take upon?

6. Did you feel any difference or challenges for ml based vs rule based?

Part3-ChatGPT

******Part3******

LLM has been identified as a promising tool for unit test generation. We want to identify if LLM can play a role in such discrimination testing. For this we considered ChatGPT.

Prompt: We provided you with a prompting template for the user to write prompts for analyzing such kinds of discrimination. Feel free to change anything in the prompting structure

Section 2: GPT Run

Scenario 1: This is a simple loan application system which is a Rule-based software for llm which is themis_llm_rulebased.py. The behavior of this software is quite deterministic. We aim to understand the user perspective on understanding discriminatory behavior of such systems through unit test generation approaches.

- **Input Specification Task**
  - ❖ For loan.py program, follow the steps one by one from chatgpt_prompt.pdf. Generate test cases and show the results.
  - ❖ For adult_income.py program, follow the steps one by one from chatgpt_prompt.pdf. Generate testcases and show the results.

- **Think-Aloud Prompt**

1. Did you face any challenges during prompt setup or required any prompt refinement?
2. What do you think these unit tests are verifying/communicating for both of the programs? Any challenges understanding them?
3. Did you face any challenge during test generation or understanding generated tests using chatgpt?

4. Do you think the gpt generated unit tests help you reason about how the program shows discrimination?

   - If so, why?
   - If not, why not?

5.Based on these gpt generated results, do you notice any potential next action you could take upon

6. Did you feel any difference or challenges for ml based vs rule based?